

GOLD COAST SEAWAY, AustraliaWMO: **945800**Lat: **27.9388S**Lon: **153.4283E**Elev: **4**StdP: **101.27**Time Zone: **10.00 (E10)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	10.8	-1.1	3.4	16.4	0.6	4.0	16.0	12.9	18.7	11.5	18.4	3.3	290	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.6	30.5	23.4	29.4	23.2	28.5	22.9	25.7	27.9	25.0	27.2	24.4	26.7	6.7	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.2	20.3	26.7	24.4	19.4	26.2	23.7	18.5	25.7	79.6	28.0	76.4	27.2	73.7	26.8	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.4	10.2	DB	7.1	34.1	1.5	1.8	6.0	35.4	5.1	36.4	4.3	37.5	3.1	38.8	(4)
(5)			WB	4.4	26.7	1.2	0.9	3.5	27.3	2.8	27.8	2.1	28.3	1.2	28.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.0	25.1	24.9	24.1	21.9	19.3	17.2	16.4	17.0	19.1	20.8	22.6	23.9	(6)	
	DBStd	3.50	1.57	1.61	1.55	1.52	1.73	1.81	1.72	1.79	1.88	1.90	1.84	1.65	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	182	0	0	0	1	11	43	64	48	12	3	0	0	(9)	
	CDD10.0	4011	467	417	437	357	287	214	198	218	274	334	377	431	(10)	
	CDD18.3	1151	209	184	178	107	39	8	3	8	36	78	127	173	(11)	
	CDH23.3	5630	1382	1201	973	342	58	6	5	22	87	222	468	864	(12)	
(13)	CDH26.7	844	249	218	129	25	1	0	0	2	8	24	57	131	(13)	
(14) Wind	WSAvg	5.0	5.6	5.7	5.6	5.1	4.5	4.4	4.3	4.6	4.9	5.3	5.2	5.3	(14)	
(15) Precipitation	PrecAvg	1522	161	180	181	143	148	85	79	63	45	105	124	150	(15)	
	PrecMax	2670	776	575	520	639	535	691	406	228	191	515	415	332	(16)	
	PrecMin	852	9	0	9	0	17	0	1	0	0	0	17	0	(17)	
	PrecStd	445	132	142	124	139	123	124	79	56	40	90	81	87	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.0	31.8	31.0	28.9	26.2	24.1	23.8	25.5	27.5	29.3	30.1	31.5	(19)	
		MCWB	24.7	24.1	23.6	21.6	19.1	17.4	16.5	16.3	17.1	18.6	20.5	21.8	(20)	
	2%	DB	30.0	30.1	29.2	27.2	24.9	22.6	22.2	23.6	25.4	26.9	28.1	29.2	(21)	
		MCWB	24.0	24.4	23.5	20.8	18.6	17.0	15.8	16.1	17.7	19.9	21.1	22.7	(22)	
	5%	DB	29.0	29.0	28.0	26.1	23.8	21.6	21.2	22.2	24.1	25.6	27.0	28.0	(23)	
		MCWB	23.7	23.7	22.9	20.6	18.2	16.4	15.6	15.7	17.9	19.7	21.1	22.4	(24)	
	10%	DB	28.1	28.0	27.1	25.2	22.9	20.7	20.2	21.1	23.0	24.4	25.9	27.1	(25)	
		MCWB	23.1	23.2	22.4	20.3	18.2	16.0	15.0	15.2	17.4	19.2	20.8	22.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.4	26.8	26.6	23.5	22.2	20.1	19.7	19.6	20.9	22.7	24.0	25.4	(27)	
		MCDB	28.3	28.8	28.3	25.9	23.3	21.6	21.2	21.9	23.3	25.0	26.3	27.4	(28)	
	2%	WB	25.5	25.8	25.4	22.6	21.1	18.9	18.5	18.4	20.1	21.7	23.1	24.5	(29)	
		MCDB	27.9	28.1	27.0	25.4	23.0	20.8	20.2	21.2	23.0	24.4	25.4	26.6	(30)	
	5%	WB	24.7	24.9	24.5	21.8	20.0	18.0	17.3	17.5	19.4	21.0	22.5	23.8	(31)	
		MCDB	27.3	27.2	26.4	24.9	22.5	20.1	19.6	20.5	22.3	23.4	25.0	26.3	(32)	
	10%	WB	24.2	24.2	23.6	21.1	19.1	17.3	16.3	16.6	18.6	20.2	21.9	23.2	(33)	
		MCDB	26.8	26.7	25.9	24.3	21.7	19.6	18.9	19.8	21.7	23.1	24.7	25.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	5.7	5.9	6.7	7.3	7.1	8.1	8.5	8.0	6.9	6.2	6.0	(35)	
		MCDBR	7.1	7.3	7.5	8.1	8.6	8.1	9.3	10.3	9.8	9.1	7.9	7.7	(36)	
	5% WB	MCWBR	3.2	3.3	3.5	3.9	4.3	4.3	4.9	5.1	4.8	4.5	3.5	3.4	(37)	
		MCWBR	6.1	6.1	6.2	6.9	6.5	6.0	7.0	8.2	7.7	7.3	6.6	6.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.390	0.377	0.360	0.340	0.320	0.315	0.305	0.317	0.360	0.384	0.382	0.396	(40)	
		taud	2.504	2.551	2.595	2.632	2.638	2.647	2.660	2.605	2.479	2.441	2.486	2.472	(41)	
	Ebn at Noon		952	948	935	908	883	866	892	918	919	931	953	949	(42)	
		Edh at Noon	115	107	97	85	76	72	74	85	107	118	116	119	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.48	5.92	5.08	4.29	3.57	2.94	3.44	4.32	5.34	5.96	6.51	6.47	(44)	
(45)	RadStd		0.52	0.47	0.29	0.21	0.20	0.20	0.27	0.33	0.42	0.51	0.51	0.67	(45)	

Historical Trends

		DBAvg	Heating		Cooling		
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